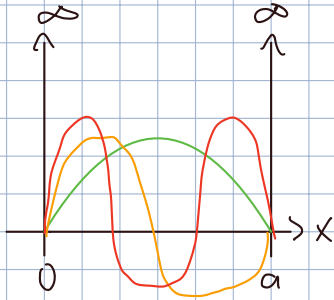


PHYS 486 - Problem Set 2

1)

ISW:

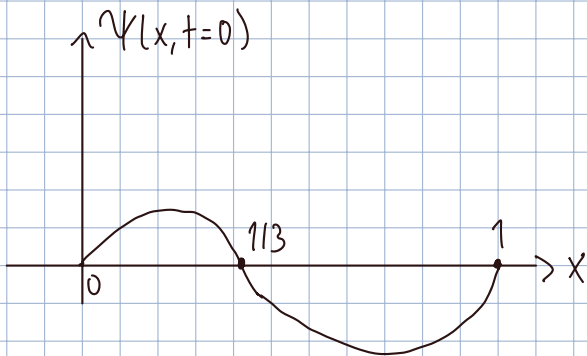


$$\Psi(x, t=0) = \begin{cases} A \left[\left(\frac{x}{a}\right)^3 - \frac{4}{3} \left(\frac{x}{a}\right)^2 + \frac{1}{3} \left(\frac{x}{a}\right) \right], & 0 < x < a \\ 0, & \text{elsewhere} \end{cases}$$

1.a) Let $\frac{x}{a} = u$ and normalize

$$\begin{aligned} \int_{-\infty}^{\infty} dx |\Psi(x, t=0)|^2 &= A^2 \int_0^1 a du \left[u^6 + \frac{16}{9} u^4 + \frac{1}{9} u^2 - \frac{8}{3} u^5 - \frac{8}{9} u^3 + \frac{2}{3} u^4 \right] \\ &= A^2 a \left[\frac{u^7}{7} + \frac{16}{45} u^5 + \frac{1}{27} u^3 - \frac{8}{18} u^6 - \frac{2}{9} u^4 + \frac{2}{15} u^4 \right]_0^1 \\ &= A^2 a \left(\frac{1}{7} + \frac{22}{45} + \frac{1-12-6}{27} \right) = A^2 a \frac{1}{9} \left[\frac{9}{7} + \frac{66}{15} - \frac{14}{3} \right] = \frac{A^2 a}{9} \left(\frac{135 + 462 - 595}{105} \right) \\ &= \frac{A^2 a}{945} \Rightarrow A = \sqrt{\frac{945}{2a}} \end{aligned}$$

$$\Psi(x, t=0) = A \left[u^3 - \frac{4}{3} u^2 + \frac{1}{3} u \right] = A u \left(u - \frac{1}{3} \right) (u - 1) \Rightarrow \text{3rd order polynomial crosses } x\text{-axis at } x=0, x=\frac{1}{3}, x=1$$



$\Rightarrow$ Normalization is required for statistical description of quantum mechanics to hold

$\rightarrow$ We require that prob. of finding the particle anywhere in space $(-\infty, \infty)$ equals to 1.

$$Pr[-\infty < x < \infty] = \int_{-\infty}^{\infty} dx |\Psi(x, t)| = 1$$

1.b) We know that we can express $\Psi(x,t)$ in terms of eigenstates of the ISW as $\Psi(x,t) = \sum_{n=1}^{\infty} c_n \Psi_n(x,t)$ since they form a complete orthonormal set and serve as a basis for the Hilbert space

From Schrödinger Eq. $i\hbar \frac{\partial \Psi_n(x,t)}{\partial t} = E_n \Psi_n(x,t)$ for each eigenstate $\Psi_n(x,t)$

Since potential is time-independent, eigenstates will be stationary:

$$\Psi_n(x,t) = \Psi_n(x) e^{-iE_n t/\hbar}$$

↳ solution from TISE

Then total wavefunction at time t will be given by:

$$\Psi(x,t) = \sum_{n=1}^{\infty} c_n \Psi_n(x) e^{-iE_n t/\hbar}$$

$$c_n = \langle \Psi(x,t=0), \Psi_n(x) \rangle = \int_0^{\infty} \Psi(x,t=0)^* \Psi_n(x) dx \quad \text{where } \Psi_n(x) = \sqrt{\frac{2}{a}} \sin\left(\frac{n\pi x}{a}\right)$$

$$= A \sqrt{\frac{2}{a}} \int_0^1 \left(u^3 - \frac{4}{3}u^2 + \frac{1}{3}u\right) \sin(n\pi u) du$$

integration by parts

$$= A \sqrt{2a} \left\{ \left[-\left(u^3 - \frac{4}{3}u^2 + \frac{1}{3}u\right) \frac{\cos(n\pi u)}{n\pi} \right]_0^1 + \frac{1}{n\pi} \int_0^1 \left(3u^2 - \frac{8u}{3} + \frac{1}{3}\right) \cos(n\pi u) du \right\}$$

$$= \frac{A \sqrt{2a}}{n\pi} \left\{ \left[\left(3u^2 - \frac{8u}{3} + \frac{1}{3}\right) \frac{\sin(n\pi u)}{n\pi} \right]_0^1 - \frac{1}{n\pi} \int_0^1 \left(6u - \frac{8}{3}\right) \sin(n\pi u) du \right\}$$

$$= -\frac{A \sqrt{2a}}{(n\pi)^2} \left\{ \left[-\left(6u - \frac{8}{3}\right) \frac{\cos(n\pi u)}{n\pi} \right]_0^1 + \frac{6}{n\pi} \int_0^1 \cos(n\pi u) du \right\}$$

$$= \frac{A \sqrt{2a}}{(n\pi)^3} \left(\frac{10}{3} (-1)^n + \frac{8}{3} \right) = \frac{3\sqrt{105}}{\sqrt{2a}} \frac{\sqrt{2a}}{(n\pi)^3} \frac{1}{3} (10(-1)^n + 8) = \frac{\sqrt{105}}{(n\pi)^3} \left[8 + 10(-1)^n \right]$$

$$c_n = \begin{cases} 18 \frac{\sqrt{105}}{(n\pi)^3} & , \quad n \text{ even} \\ -2 \frac{\sqrt{105}}{(n\pi)^3} & , \quad n \text{ odd} \end{cases}$$

$$\Psi(x,t) = \sum_{n=1}^{\infty} c_n \Psi_n(x) e^{-iE_n t/\hbar} \Rightarrow \text{express i.t.o. ISW energy eigenstates}$$

$$\Psi(x,t) = \sum_{n=1}^{\infty} \frac{\sqrt{105}}{(n\pi)^3} \left[8 + 10(-1)^n \right] \sqrt{\frac{2}{a}} \sin\left(\frac{n\pi x}{a}\right) \exp\left[\frac{-i\hbar \pi^2 n^2}{2ma^2} t\right]$$

1.c) Probability of measuring an eigenvalue E_n is $|c_n|^2$

$$P(E_n) = |c_n|^2$$

* As a side note $\sum_{n=1}^{\infty} P(E_n) = \sum_{n=1}^{\infty} |c_n|^2 = \int \Psi^*(x,t) \Psi(x,t) dx = 1$

Thus, total prob. is $\sum_{n,m} c_n^* c_m \int \Psi_n^* \Psi_m dx = \sum_n |c_n|^2 = 1$, in conformity.

$$P(E_n) = \frac{105}{(n\pi)^6} |8 + 10(-1)^n|^2 = \begin{cases} \frac{34020}{(n\pi)^6}, & n \text{ even} \\ \frac{420}{(n\pi)^6}, & n \text{ odd} \end{cases}$$

1.d) $\langle E \rangle = \sum_n E_n P(E_n)$ OR $\langle E \rangle = \langle \Psi | \hat{H} | \Psi \rangle$

$$\begin{aligned} \langle E \rangle &= \sum_n \frac{\hbar^2 \pi^2 n^2}{2ma^2} \left(\frac{105}{n^6 \pi^6} \right) |8 + 10(-1)^n|^2 \\ &= \frac{105 \hbar^2}{2ma^2 \pi^4} \left[4 \sum_{n \text{ odd}} \frac{1}{n^4} + 324 \sum_{n \text{ even}} \frac{1}{n^4} \right] \end{aligned}$$

Using the fact: $\sum_{n=1}^{\infty} \frac{1}{n^4} = \zeta(4) = \frac{\pi^4}{90}$ (Riemann Zeta function)

$$\sum_{n \text{ even}} \frac{1}{n^4} = \sum_{k=1}^{\infty} \frac{1}{(2k)^4} = \frac{1}{16} \times \frac{\pi^2}{90} \Rightarrow \sum_{n \text{ odd}} \frac{1}{n^4} = \sum_{n=1}^{\infty} \frac{1}{n^4} - \sum_{n \text{ even}} \frac{1}{n^4} = \frac{15}{16} \times \frac{\pi^2}{90}$$

$$\langle E \rangle = \frac{105 \hbar^2}{2m a^2 \pi^4} \frac{\pi^4}{90} \left(\underbrace{4 \times \frac{15}{16} + 324 \times \frac{1}{16}}_{\frac{96}{4}} \right) = \boxed{\frac{14 \hbar^2}{m a^2}}$$

→ Prob. for energy measurement to return to ground state $E_1 = \frac{\hbar^2 \pi^2}{2m a^2}$ is $|c_1|^2$ which is indep of time.

→ $\langle E \rangle$ also doesn't depend on time since neither $|c_n|^2$ nor E_n depends on time.

↳ Also $\langle E \rangle = \langle \Psi | \hat{H} | \Psi \rangle = \int_0^a \sum_n c_n \psi_n^* e^{iE_n t/\hbar} \sum_m c_m E_m \psi_m e^{-iE_m t/\hbar} dx$

$$= \sum_{n,m} c_n^* c_m E_n e^{i(E_n - E_m)t/\hbar} \delta_{nm} = \sum_n |c_n|^2 E_n$$

removes time dependence

* Hamiltonian is not time dep. and energy is conserved.

$$2.a) \quad -\frac{\hbar^2}{2m} \frac{\partial^2 \psi_n(x)}{\partial x^2} + V(x) \psi_n(x) = E_n \psi_n(x) \quad \times \text{multiply by } \psi_m$$

$$\text{Subtract} \quad -\frac{\hbar^2}{2m} \frac{\partial^2 \psi_m(x)}{\partial x^2} + V(x) \psi_m(x) = E_m \psi_m(x) \quad \times \text{multiply by } \psi_n$$

$$-\frac{\hbar^2}{2m} (\psi_n'' \psi_m - \psi_m'' \psi_n) = (E_n - E_m) \psi_n \psi_m$$

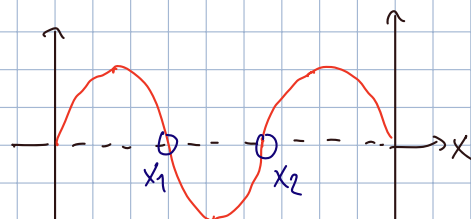
Note that $\frac{d}{dx} (\psi_n' \psi_m - \psi_m' \psi_n) = \psi_n'' \psi_m + \psi_n' \psi_m' - \psi_m'' \psi_n - \psi_m' \psi_n'$
 $= \psi_n'' \psi_m - \psi_m'' \psi_n$

beware of (-) sign

$$\frac{d}{dx} \left(\frac{d\psi_m}{dx} \psi_n - \psi_m \frac{d\psi_n}{dx} \right) = \frac{2m}{\hbar^2} (E_n - E_m) \psi_n \psi_m$$

$$2.b) \quad \text{Note that at nodes of } \psi_m, \psi_m(x_1) = \psi_m(x_2) = 0$$

Integrating the previously found equation:



$$\left[\psi_m' \psi_n - \psi_m \psi_n' \right]_{x_1}^{x_2} = \frac{2m}{\hbar^2} (E_n - E_m) \int_{x_1}^{x_2} \psi_n \psi_m dx$$

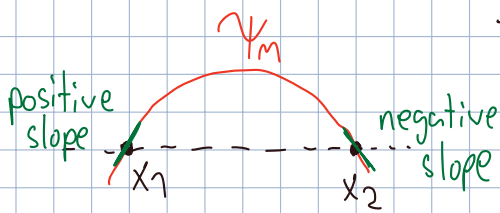
$$\psi_m'(x_2) \psi_n(x_2) - \cancel{\psi_m(x_2)} \psi_n'(x_2) - \psi_m'(x_1) \psi_n(x_1) + \cancel{\psi_m(x_1)} \psi_n'(x_1)$$

$$= \boxed{\psi_m'(x_2) \psi_n(x_2) - \psi_m'(x_1) \psi_n(x_1) = \frac{2m}{\hbar^2} (E_n - E_m) \int_{x_1}^{x_2} \psi_n \psi_m dx}$$

2.c)

Assume that $\Psi_m > 0$ in $x_1 < x < x_2$ (with $x_1 < x_2$ of course)

→ clearly $\Psi'_m(x_1) > 0$, $\Psi'_m(x_2) < 0$



→ LHS of eq. from part (b) claims then

$$\text{sign} \left[\underbrace{\Psi'_m(x_2)}_{<0} \Psi_n(x_2) - \underbrace{\Psi'_m(x_1)}_{>0} \Psi_n(x_1) \right] = -\text{sign}[\Psi_n], \quad x \in [x_1, x_2]$$

→ RHS on the other claims

$$\text{sign} \left[\frac{2m}{\hbar^2} (E_n - E_m) \int_{x_1}^{x_2} dx \underbrace{\Psi_m \Psi_n}_{<0} \right] = \text{sign}[\Psi_n] \quad \text{since } \Psi_n \text{ has the same sign throughout } [x_1, x_2]$$

∴ There exist a contradiction about LHS and RHS of eq. claiming different signs

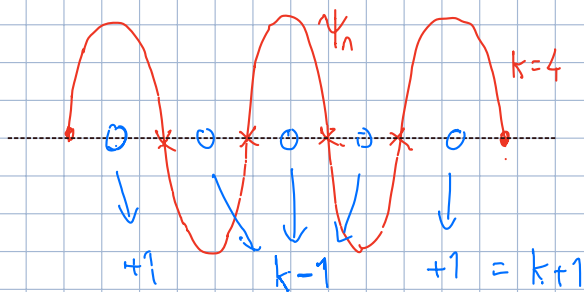
→ Similarly for $\Psi_m < 0$, LHS claims $\text{sign}[\Psi_n]$ whereas RHS claims $-\text{sign}[\Psi_n]$

⇒ Thus there must exist at least one node of $\Psi_n(x)$ between $x_1 < x < x_2$.

* In fact Ψ_n will have more nodes than Ψ_m .

↳ If Ψ_n has k nodes, Ψ_m will have $k-1$ nodes between Ψ_n 's

nodes + 2 nodes between endpoints and left/rightmost nodes



3) $\psi_n(x) = \sqrt{\frac{2}{L}} \sin\left(\frac{n\pi x}{L}\right)$: n^{th} stationary state of ISW with length L

$$\begin{aligned} \langle x \rangle &= \langle \psi_n | x | \psi_n \rangle = \int_0^L \psi_n^*(x) \hat{x} \psi_n(x) dx = \frac{2}{L} \int_0^L x \sin^2\left(\frac{n\pi x}{L}\right) dx \\ &= \frac{2}{L} \int_0^L \frac{x}{2} \left[1 - \cos\left(\frac{2n\pi x}{L}\right) \right] dx = \frac{1}{L} \int_0^L x dx - \frac{1}{L} \int_0^L x \cos\left(\frac{2n\pi x}{L}\right) dx \\ &= \frac{1}{L} \left. \frac{x^2}{2} \right|_0^L = \boxed{\frac{L}{2}} \end{aligned}$$

$\left[\frac{xL}{2\pi n} \sin\left(\frac{2\pi n x}{L}\right) \right]_0^L - \frac{L}{2\pi n} \int_0^L \sin\left(\frac{2\pi n x}{L}\right) dx = 0$

$$\begin{aligned} \langle x^2 \rangle &= \frac{2}{L} \int_0^L x^2 \sin^2\left(\frac{n\pi x}{L}\right) dx = \frac{1}{L} \int_0^L x^2 dx - \frac{1}{L} \int_0^L x^2 \cos\left(\frac{2n\pi x}{L}\right) dx \\ &= \frac{L^2}{3} - \frac{1}{L} \left\{ \left[\frac{Lx^2}{2\pi n} \sin\left(\frac{2\pi n x}{L}\right) \right]_0^L - \frac{2L}{2\pi n} \int_0^L x \sin\left(\frac{2\pi n x}{L}\right) dx \right\} \\ &= \frac{L^2}{3} + \frac{1}{\pi n} \left[- \left[\frac{xL}{2\pi n} \cos\left(\frac{2\pi n x}{L}\right) \right]_0^L + \frac{L}{2\pi n} \int_0^L \cos\left(\frac{2\pi n x}{L}\right) dx \right] = \frac{L^2}{3} - \frac{L^2}{2(\pi n)^2} \end{aligned}$$

$$\boxed{\langle x^2 \rangle = L^2 \left(\frac{1}{3} - \frac{1}{2(n\pi)^2} \right)}$$

$$\begin{aligned} \langle p \rangle &= \int_0^L \sqrt{\frac{2}{L}} \sin\left(\frac{n\pi x}{L}\right) (-i\hbar) \frac{\partial}{\partial x} \left(\sqrt{\frac{2}{L}} \sin\left(\frac{n\pi x}{L}\right) \right) dx \\ &= -\frac{2i\hbar}{L} \cdot \frac{1}{n\pi} \int_0^L \sin\left(\frac{n\pi x}{L}\right) \cos\left(\frac{n\pi x}{L}\right) dx = \frac{-i\hbar}{n\pi} \int_0^L \sin\left(\frac{2n\pi x}{L}\right) dx = \boxed{0} \end{aligned}$$

$$\begin{aligned}
 \langle p^2 \rangle &= \int \sqrt{\frac{2}{L}} \sin\left(\frac{n\pi x}{L}\right) (-i\hbar)^2 \frac{d^2}{dx^2} \left[\sqrt{\frac{2}{L}} \sin\left(\frac{n\pi x}{L}\right) \right] dx = \frac{+2\hbar^2}{L} \left(\frac{n\pi}{L}\right)^2 \int_0^L \sin^2\left(\frac{n\pi x}{L}\right) dx \\
 &= \frac{2\hbar^2 \pi^2 n^2}{L^3} \left[\int_0^L \frac{1}{2} dx - \frac{1}{2} \int_0^L \cos \frac{2n\pi x}{L} dx \right] = \boxed{\left(\frac{\hbar \pi n}{L}\right)^2}
 \end{aligned}$$

$$\sigma_x = \sqrt{\langle x^2 \rangle - \langle x \rangle^2} = L \sqrt{\frac{1}{12} - \frac{1}{2(n\pi)^2}}, \quad \sigma_p = \sqrt{\langle p^2 \rangle} = \frac{\hbar \pi n}{L}$$

$$\sigma_x \sigma_p = \hbar \pi n \sqrt{\frac{1}{12} - \frac{1}{2(n\pi)^2}} = \frac{\hbar}{2} \sqrt{\frac{n^2 \pi^2}{3} - 2}$$

$$\text{For } n=1 \quad \sqrt{\frac{\pi^2}{3} - 2} \approx 1.136 \Rightarrow \text{Thus } \sigma_x \sigma_p \geq \frac{\hbar}{2}$$

$$\hookrightarrow \text{Therefore } \sigma_x \sigma_p \geq \frac{\hbar}{2}, \quad \forall n \in \{1, 2, 3, \dots\}$$

$$\text{since } \left(\frac{n^2 \pi^2}{3} - 1\right)^{1/2} \text{ is an increasing func.}$$

$$4) \quad \Psi_A(x, t=0) = \sqrt{\frac{1}{2}} \Psi_1(x) + \sqrt{\frac{1}{6}} \Psi_2(x) + \sqrt{\frac{1}{3}} \Psi_3(x)$$

where $\Psi_i(x)$ denotes i^{th} stationary eigenstate of ISW.

$$4.a) \quad \int_{-\infty}^{\infty} \Psi_A^*(x, t) \Psi_A(x, t) dx = \frac{1}{2} |\Psi_1|^2 + \frac{1}{6} |\Psi_2|^2 + \frac{1}{3} |\Psi_3|^2 + \sqrt{\frac{1}{18}} \int_{-\infty}^{\infty} (\cancel{\Psi_2^* \Psi_3} + \cancel{\Psi_3^* \Psi_2}) dx \\ + \int_{-\infty}^{\infty} \sqrt{\frac{1}{12}} (\cancel{\Psi_1^* \Psi_2} + \cancel{\Psi_2^* \Psi_1}) + \sqrt{\frac{1}{6}} \int_{-\infty}^{\infty} (\cancel{\Psi_1^* \Psi_3} + \cancel{\Psi_3^* \Psi_1}) dx$$

Recall that eigenfunctions form an orthonormal set:

$$\int_{-\infty}^{\infty} \Psi_n \Psi_m dx = \delta_{nm}$$

That is why all the cross-terms are zero and $|\Psi_i|^2 = 1$

$$\text{Then } \int \Psi_A^*(x, t=0) \Psi_A(x, t=0) dx = \frac{1}{2} + \frac{1}{3} + \frac{1}{6} = 1$$

To construct $\Psi_B(x, t=0)$ with same energies and probs.:

↳ we can't change states Ψ_1, Ψ_2, Ψ_3 in its composition

↳ we must still have:

$$\left. \begin{aligned} |C_1|^2 &= |C_1'|^2 = \frac{1}{2} \\ |C_2|^2 &= |C_2'|^2 = \frac{1}{6} \\ |C_3|^2 &= |C_3'|^2 = \frac{1}{3} \end{aligned} \right\} \begin{array}{l} \text{primed } C_n \text{ are} \\ \text{new coeffs of states} \end{array}$$

* We can have $C_1 = \pm \frac{1}{2}$, $C_2 = \pm \frac{1}{6}$, $C_3 = \pm \frac{1}{3}$ except all C_i 's are positive or all negative. One possible Ψ_B is

$$\Psi_B(x, t=0) = \frac{1}{\sqrt{2}} \Psi_1(x) - \frac{1}{\sqrt{6}} \Psi_2(x) - \frac{1}{\sqrt{3}} \Psi_3(x) \quad ,$$

4.b)

$$\begin{aligned}\langle H \rangle &= \sum_n P(n) E_n = \sum_n |c_n|^2 \frac{\hbar^2 \pi^2 n^2}{2mL^2} \\ &= \frac{\hbar^2 \pi^2}{2mL^2} \left(\underbrace{\frac{1}{2} \cdot 1 + \frac{1}{6} \cdot 4 + \frac{1}{3} \cdot 9}_{\frac{25}{6}} \right) = \frac{25 \hbar^2 \pi^2}{12 mL^2}\end{aligned}$$

We need to pick some states with

$$\sum_n |c_n|^2 = 1 \quad \text{and} \quad \sum_n |c_n|^2 n^2 = \frac{25}{6}$$

Let us have 2 states; ψ_1 and ψ_4 for example

$$\left. \begin{aligned} |c_1|^2 + |c_4|^2 &= 1 \\ 1|c_1|^2 + 16|c_4|^2 &= \frac{25}{6} \end{aligned} \right\} \quad \begin{aligned} 1 + 15|c_4|^2 &= \frac{25}{6} \Rightarrow |c_4|^2 = \frac{19}{90} \\ |c_1|^2 &= 1 - |c_4|^2 = \frac{71}{90} \end{aligned}$$

A possible ψ_c is

$$\psi_c(x, t=0) = \sqrt{\frac{71}{90}} \psi_1(x) + \sqrt{\frac{19}{90}} \psi_4(x)$$

$\hookrightarrow \psi_c$ has a different set of measured energies $\Rightarrow E_1, E_4$

$\hookrightarrow \psi_c$ is linearly indep. of ψ_B since $\psi_i(x)$'s are orthogonal

Check that $\langle \psi_c | \hat{H} | \psi_c \rangle = \frac{71}{90} + \frac{19}{90} \times 16 = \frac{375}{90} = \frac{25}{6}$ //